

Grain structure and texture evolution across the deflected laser welded Fe-Cu dissimilar joint

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ABSTRACT

In this paper, a deflected laser welded joint of 304 stainless steel and T2 copper was studied concerning structural arrangement, grain structure and texture evolution. According to SEM and EBSD results, the as-welded fusion zone contained a major fraction of Fe-Cr-Ni rich α structure and minor fraction of Cu-rich ϵ structure. Furthermore, three types of ϵ were observed including blocky, curved and spherical forms. Both grain size and crystalline orientation varied largely from 304 side to T2 side. Additionally, epitaxial zone was progressively wider on 304 side than T2. In mixed zone, ϵ possessed comparably dispersed orientation and fine grains. Based on solidification sequence of $\delta \rightarrow$ Fe-Cr-Ni rich $\delta/\gamma \rightarrow$ Cu rich γ in the fusion zone, the dispersed primarily solidified of Fe-Cr-Ni rich spheres with random orientation led to the formation of fine grains in ϵ . As for α , the continuous growth of Fe-Cr-Ni rich bulks with uniform orientation promoted the identical growth and orientation of inside Cu rich spheres.

1. Introduction

Dissimilar fusion welded joints were frequently encountered when two kinds of materials with distinct element, lattice and physicochemical properties [1–3]. According to the reaction features, such joints were divided into two types. One was called the brittle joint generally containing intermetallic compounds (IMCs), such as Ti-Al [4], Mg-Al [5,6] and Fe-Al [7,8], etc. The other kind of solid-solution joints were free of IMCs including Fe-Cu [9], Fe-Ni [10] and other coessential materials with varied compositions [11,12]. From a welding view-of-point, the dissimilar material joints were supposed to first experience melting, then mixing and finally solidification. In this way, the formation of IMCs or solid-solution structures were assumed to occur in the terminal solidification. Moreover, such formation process was dominantly influenced by the melting and mixing stages. Recently, laser welding has become a high efficiency-quality joining technology. During laser welding of dissimilar materials, violent pool stirring, rapid solidification speed and high temperature gradient resultantly produced miscible and heterogeneous as-weld structures in room temperature. Here, Fe-Cu laser welded joint was chosen here for clarifying the melting, mixing and solidification features for solid-solution type joints.

So far, there have been many studies for laser welded Fe-Cu joints. In the welding configuration, it was found that a laser deflection of 6° to 8° was beneficial to a sound weld free of microcracks [13]. After welding process, the weld bead generally featured an asymmetric formation in the transverse section [14]. By computational simulation, such appearance was immediately attributed to the heterogeneous temperature profiles and flow velocity on Cu and steel sides [15]. At the meantime, the microstructure and property presented a complicated and inconsistent variation from Cu side to Fe side [8,16–19]. From a metallurgical point of view, the fraction of Cu phase was necessarily controlled considering the microcracks induced by thermal stress mismatch in fusion zone [20]. In heat affected zone, liquation cracks were also observed via crack incubation, initiation and growth, primarily resulting from the overloaded heat input in such a narrow weld [21]. Therefore, the laser power was supposed to limit within 3000 W.

Considering such features, the main purpose of this study was to clarify the effect of discrepancy in melting point on the weld as-solidified structures. First, the overall morphology and microhardness of weld bead in the transverse section was presented. Then, primary microstructure features in fusion zone and relevant evolution of crystalline orientation and grain structure across the joint was studied.

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Finally, the formation process of typical structure and texture in fusion zone was illuminated.

2. Experiment

In this study, 100 mm × 50 mm × 2 mm plates of 304 type austenite stainless steel (304) and T2 type copper (T2) were utilized here to serve as butt welding materials. The nominal chemical compositions of two base metal (BM) were listed in Table 1.

The heat source was generated by IPG YLR-4000, a single-mode continuous-wave laser focusing as a spot of 0.5 mm. To avoid the “high-reflection” phenomenon and relatively effective fusion effect, laser incidence was located at the 304/T2 interface with no defocusing but angled at 10° from the vertical line to 304 side, as shown in Fig. 1. A three-dimensional coordinate system was established: rolling direction (RD), transverse direction (TD) and normal direction (ND). The laser power and welding speed were set as 3000 W and 2.4 m/min. Meanwhile, the shielding gas was a pure argon with a flow velocity of 1.5 m³/h.

The typical sample of the welded joints was cut from transverse sections (RD-TD plane). After grinding and polishing, the sample was etched in a solution of 40 mL H₂O + 30 mL HNO₃ + 10 mL HCl and subsequently recorded in back scattered electron (BSE) mode to determine macro- and micro- structures [22]. After ion etching treatment, EBSD test was performed to further investigate the evolution of texture and grain structure across the joint. For simplicity, face-centered cube whose $\alpha/\beta/\gamma = 90^\circ$ and $a/b/c = 2.866 \text{ \AA}$, were fixed as the single crystal structure of 304 and Cu. Vickers hardness mapping was conducted on the surface of transversal section to illustrate the variation of microstructure.

3. Results and discussion

In Fig. 2a, a deflected fusion zone (FZ) was observed with polymorphic white bands arranged in the matrix. Overall, the fraction of ϵ was comparably lower than that of α , primarily resulting from the better absorptivity of 304 than T2. The former was relatively close to T2 BM while the latter the 304 BM, indicating a Cu rich ϵ structure and Fe-Cr-Ni rich α structure. In the weld top, ϵ presented a blocky shape on T2 side and a spherical type on 304 side [23]. In the weld bottom, ϵ exhibited a vineous shape on T2 side and a curved one on 304 side. Therefore, weld pool experienced a comparably violent fluctuation and Marangoni convection on 304 side [24]. According to the mapping in Fig. 2b, the average hardness value of FZ was located between T2 BM and 304 BM. Furthermore, ϵ displayed a lower value than α . Hence, the FZ of T2 and 304 acted as a naturally transition region between hard and soft material.

In an amplified scale, on the T2 side (Fig. 2c), there existed a mixture of pure Cu and blocky ϵ with a zigzag fusion line (FL). In such region, α was observed in a progressively fine and spherical fashion, indicating a higher fraction of Cu than Fe from a viewpoint of Fe-Cu alloy. Far from this region, ϵ and α appeared alternately. Therefore, fine ϵ in α and fine α in ϵ were both observed. Fig. 2d, e and f displayed three regions featured with bonded, curved and spherical ϵ . Overall, the α inside exhibited a higher dispersed form in the bonded than the curved, in return suggesting the linear and annular convection in the weld pool. The discrete feature of ϵ revealed the strongest unstable flow at the top of 304 side. Different from the T2 side, there presented a protrusion of pure BM on 304 side, mainly resulting from the quick solidification of 304 melt in an

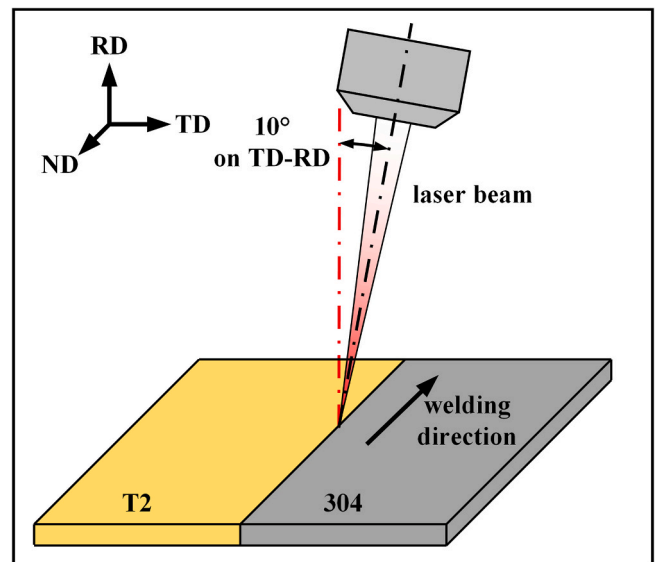


Fig. 1. Schematic diagram for laser welding T2 to 304.

undercooling condition [22].

To clarify the evolution of grain structure and texture, EBSD test and relevant posttreatment were conducted from 304 BM to T2 BM on the middle penetration, as shown in Fig. 3. Fusion line (FL) was definitely distinguished depending on the growth and morphology of grains. In Fig. 3a, the variation of grain morphology indicated seven subregions including 304 BM, epitaxial, mixed zone, epitaxial, heat affected zone (HAZ) and T2 BM. On the 304 side, there presented an grain coarsening trend from BM to HAZ and epitaxial zone. On the T2 side, however, grain size first increased in HAZ and then refined in epitaxial zone. Furthermore, T2 side displayed a wider HAZ but narrower epitaxial zone than 304 side. In the mixed zone, fine and coarse grains were simultaneously encountered in the mixed zone. In Fig. 3b, there existed grain coarsening in both HAZs of 304 and T2 sides. Compared to ϵ , primarily solidified α took on a coarse and sustaining fashion in epitaxial zones instead. As can be inferred from Fig. 3c and d, the presence of ϵ was accompanied by fine grains and dispersed orientation while α the coarse and uniform forms. Therefore, it was reasonable to confirm the huge difference in the solidification of α and ϵ structures.

Fig. 4 exhibited the distribution of grain size and crystalline orientation on 304 side. Note the maximum grain size in HAZ (51 μm) was relatively higher than that in BM (48 μm), resulting in an average of 17 μm . In addition, the dispersed distribution of orientation in BM and HAZ induced concurrent low pole densities of 2.68 and 3.33, respectively. This appearance indicated that grain coarsening in HAZ brought out negligible changes from BM. Such appearance verified the typical equiaxed feature after annealing and solution treatment.

In the epitaxial zone of 304 side, the major fashion was presented with continuous growth and uniform orientation, as displayed in Fig. 5. At the initial stage, a narrow-width zone was established with fine grains and dispersed orientations. Subsequently, crystalline orientation parallel to heat diffusion was chosen out and promoted. Resultantly, a large grain was observed with a size of 230 μm . Textually, the growth direction was primarily located at $[-103]$ and $[311]$ crystalline directions with a high pole density of 14.4. In addition, $[-103]$ point conformed immediately to the welding direction.

Fig. 6 exhibited the distribution of grain size and crystalline orientation on T2 side. Apparently, the maximum grain size in HAZ (106 μm) was relatively higher than that in BM (67 μm), resulting in an average of 18 μm . As a result, like the 304 side, a similar appearance of dispersed orientation in HAZ and BM was observed with low pole densities of 2.61 and 3.47, respectively. Meantime, grain coarsening on Cu side

Table 1

Chemical composition of the 304 and T2 (wt%).

Materials	Cu + Ag	Fe	Cr	Ni	Mn	Si
304	–	Bal	19.21	9.13	1.56	0.87
T2	99.92	–	–	–	–	–

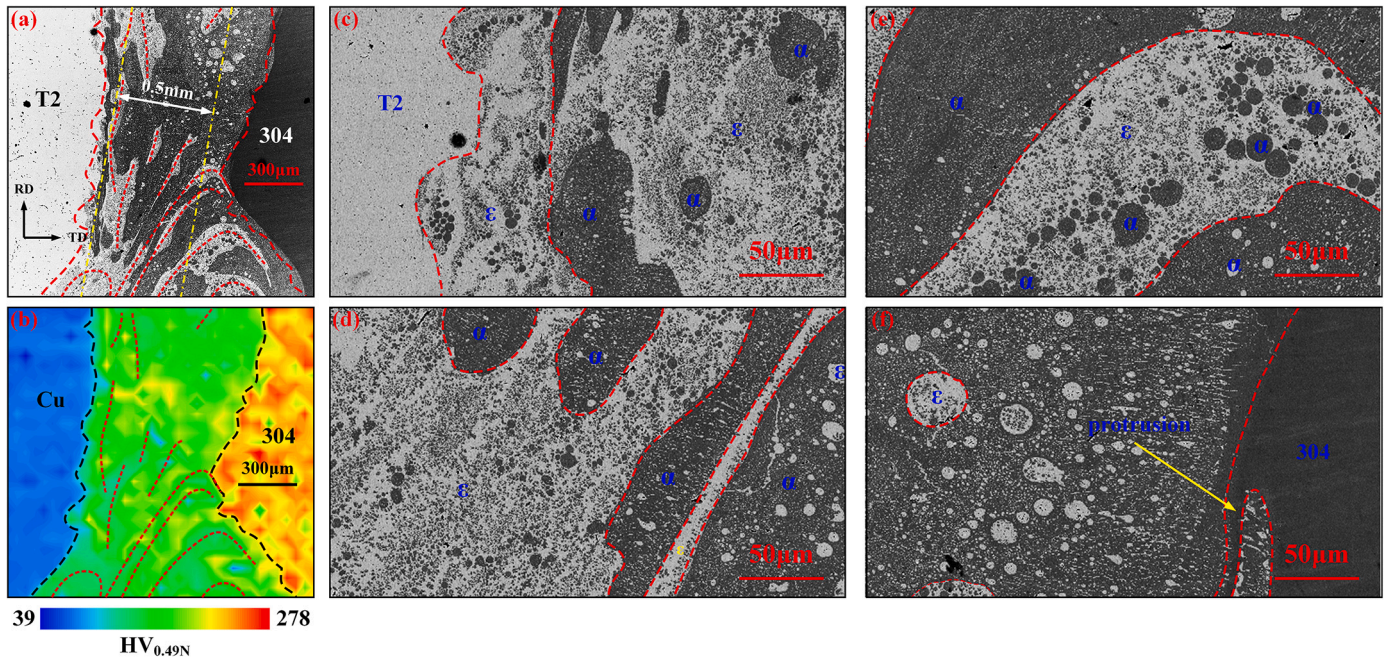


Fig. 2. Microstructure characteristics of ϵ structure across the joint. (a) weld formation and structures; (b) microhardness mapping of the dissimilar joint; (c) blocky type on T2 side; (d) banded type; (e) curved type; (f) spherical type on 304 side.

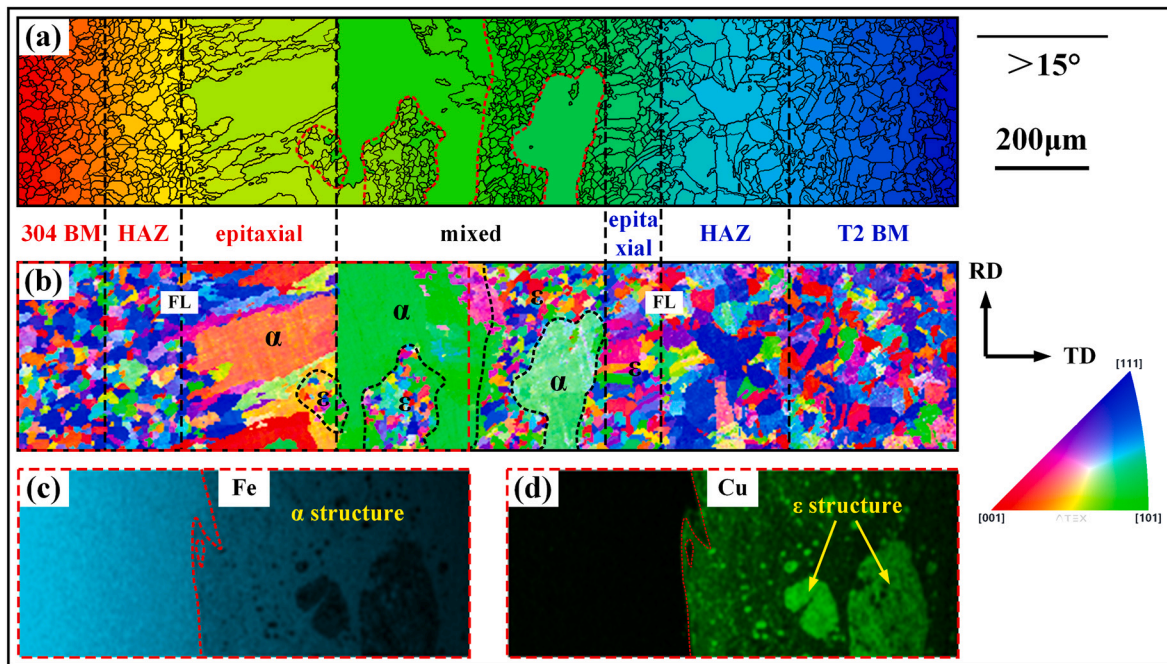


Fig. 3. Evolution of grain structure and crystalline orientation. (a) grain boundary map with 15° disorientation; (b) IPF orientation map derived from TD axis; (c) and (d) element mappings of Fe and Cu on 304 sides.

presented nearly few changes. However, compared to 304 side, the coarsening phenomenon was more obvious in T2 side. Such difference was mainly attributed the existence of δ phase, therefore inhibiting the overgrowth of grains on 304 side.

In the epitaxial zone of T2 side, there presented discontinuous growth and scattered orientation from FL to mixed zone, as exhibited in Fig. 7. Resultantly, grains were found with a maximum and average size of 68 μm and 14 μm . In such a narrow region, the growth direction was primarily located at $[-5\ 7\ 3]$ and $[0\ -2\ 7]$ crystalline directions with a comparably low pole density of 3.9. Likewise, $[0\ -2\ 7]$ point was close to

welding direction while $[-5\ 7\ 3]$ point was originated from $[-232]$ point in BM and HAZ. Compared to the epitaxial zone of 304 side, T2 side presented a weak growth continuity with dispersed crystal orientations.

Finally, consider the mixed zone in Fig. 8. Huge grains were confronted in α structure with a maximum of 323 μm while tiny grains in ϵ structure. As the primary precipitated phase, α seemed to be more easier controlled by optimal temperature gradient along TD axis, textually relating to $[6\ -1\ 1]$ crystalline orientation with a high pole density of 10.3. For ϵ structure, it exhibited a progressively fine grain

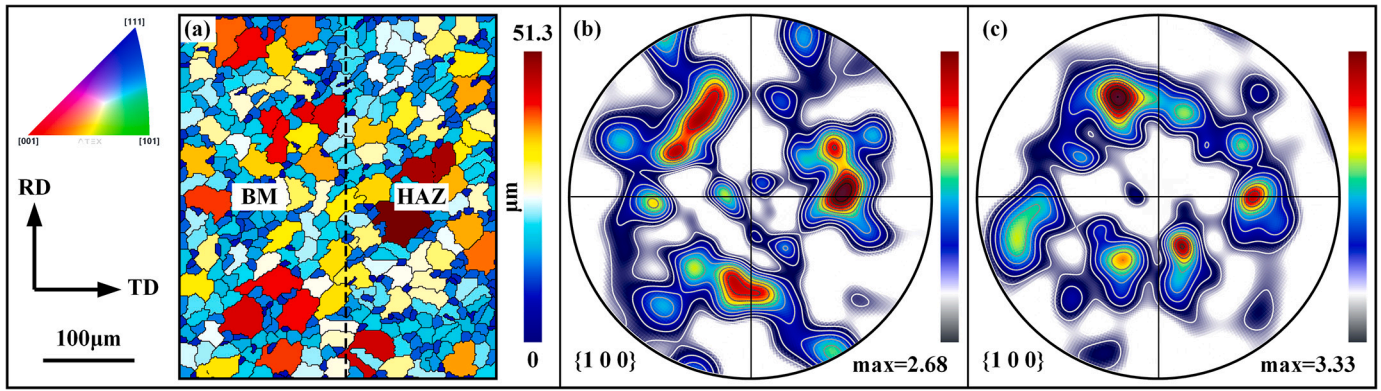


Fig. 4. Dimension and crystalline features on 304 side. (a) grain size map; (b) $\{100\}$ pole figure of BM; (c) $\{100\}$ pole figure of HAZ.

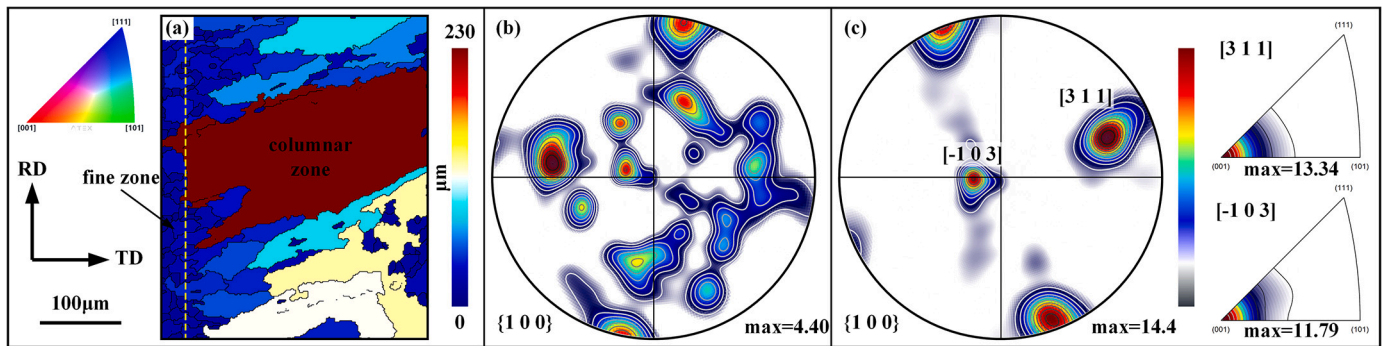


Fig. 5. Dimension and crystalline features on epitaxial growth from 304 BM. (a) grain size map; (b) $\{100\}$ pole figure of fine zone; (c) $\{100\}$ pole figure and inverse pole figures derived from $[-1\ 0\ 3]$ and $[3\ 1\ 1]$ pole points of columnar zone.

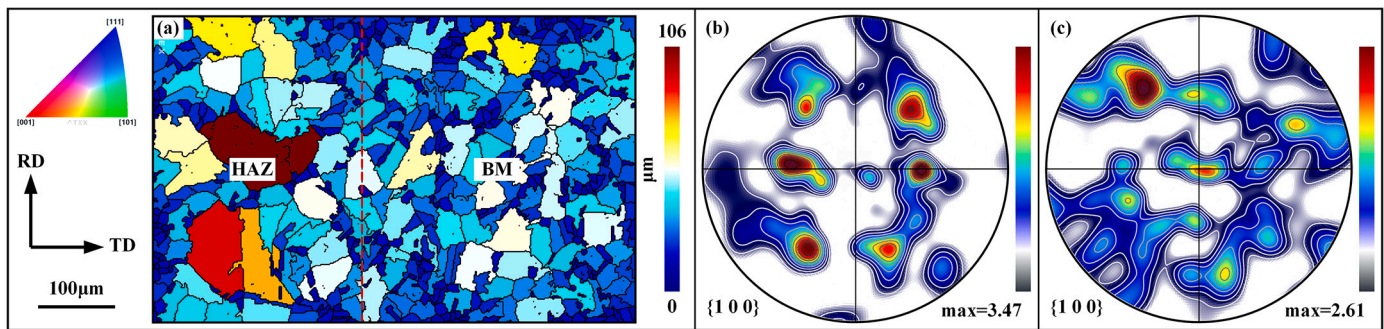


Fig. 6. Dimension and crystalline features on T2 side. (a) grain size map; (b) $\{100\}$ pole figure of HAZ; (c) $\{100\}$ pole figure of BM.

arrangement without apparent crystalline orientation concentration. Such appearance suggested two differed grain distribution and growth features.

On 304 side, the epitaxial growth with grain coarsening was a typical and conventional appearance [25–27]. To investigate the origin of such tremendous difference of grain size and distribution, further studies were conducted on the epitaxial zone on T2 side and the mixed zone inside, as shown in Fig. 9. In the EDS mappings of Fe element, there existed three kinds of arrangements. In the epitaxial zone, Fe was displayed in a dispersed and banded (Fig. 9a). In the mixed zone, Fe acted as the matrix of α while the spherical and dotted fashion in ϵ (Fig. 9e). Furthermore, in the phase maps (Fig. 9b and f), δ fraction was relatively lower in epitaxial zone (0.61%) than that in the mixed one (2.18%). In view of the indispensable role of Cr atoms in forming δ phase, a longer distance from 304 FL to T2 FL reduced the diffusion or migration of Cr. In addition, δ phase was more frequently observed in ϵ than α , probably

resulting from the solid-state transformation of $\delta \rightarrow \gamma$ in α via redistribution of Ni and Cr.

As exhibited in Fig. 9c and d, on the T2 side, the presence of δ and Fe-Cr-Ni rich γ phase blocked the conventionally epitaxial growth from T2 HAZ and gave rise to refined Fe-Cr-Ni rich and Cu rich γ grains in ϵ . This demonstrated that the a dispersed distribution of Fe-Cr-Ni rich phase in Cu-matrix exerted a heterogeneous nucleation effect. In the mixed zone (Fig. 9g and h), another situation occurred. In the subregion of α , adjacent Fe-Cr-Ni rich and Cu rich γ phase possessed the same orientation, textually leading to the formation of only several orientation outcomes. This suggested the growth dependency of Cu rich γ grains on Fe-Cr-Ni rich ones. For ϵ subregion, such phenomenon was also confronted seeing the succeeding growth from α to ϵ at their demarcation. Additionally, the dispersed orientation of δ phase was accompanied by same distribution of Fe-Cr-Ni rich and Cu rich γ phase.

Fig. 10a and b plotted the phase diagrams for Fe-Cu and Cr-Ni at 70%

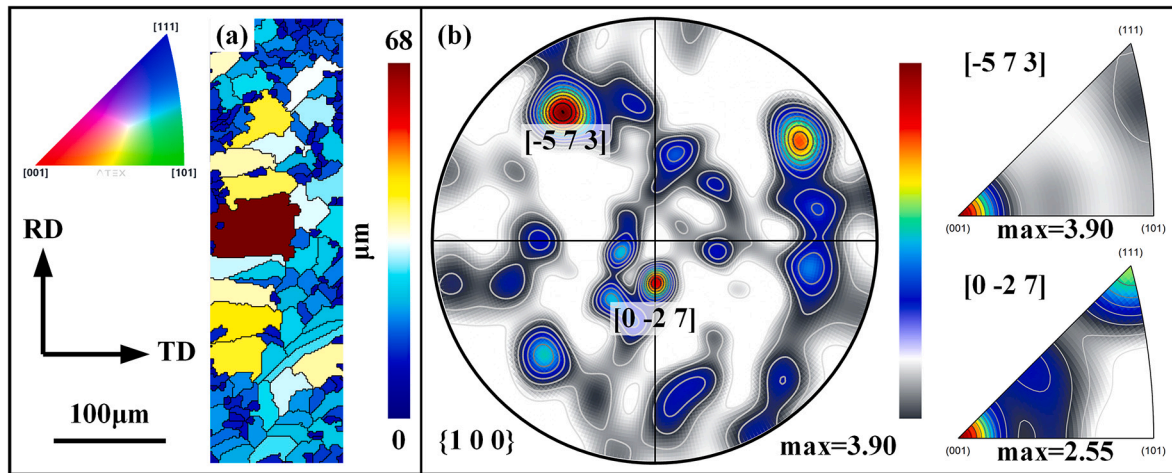


Fig. 7. Dimension and crystalline features at epitaxial zone from T2 side. (a) grain size map; (b) $\{1\ 0\ 0\}$ pole figure and inverse pole figures derived from $[-5\ 7\ 3]$ and $[0\ -2\ 7]$ pole points.

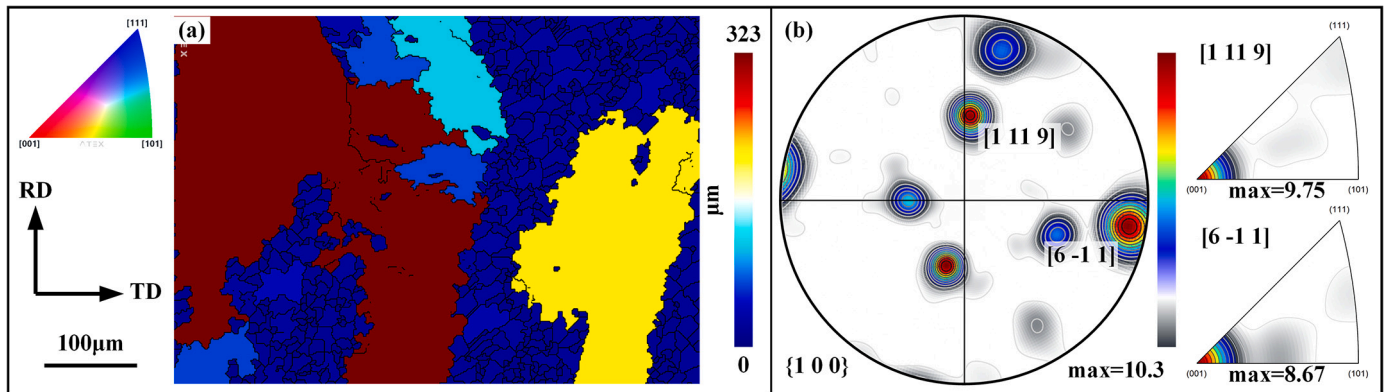


Fig. 8. Dimension and crystalline features at mixed zone. (a) grain size map; (b) $\{1\ 0\ 0\}$ pole figure and inverse pole figures derived from $[1\ 11\ 9]$ and $[6\ -1\ 1]$ pole points.

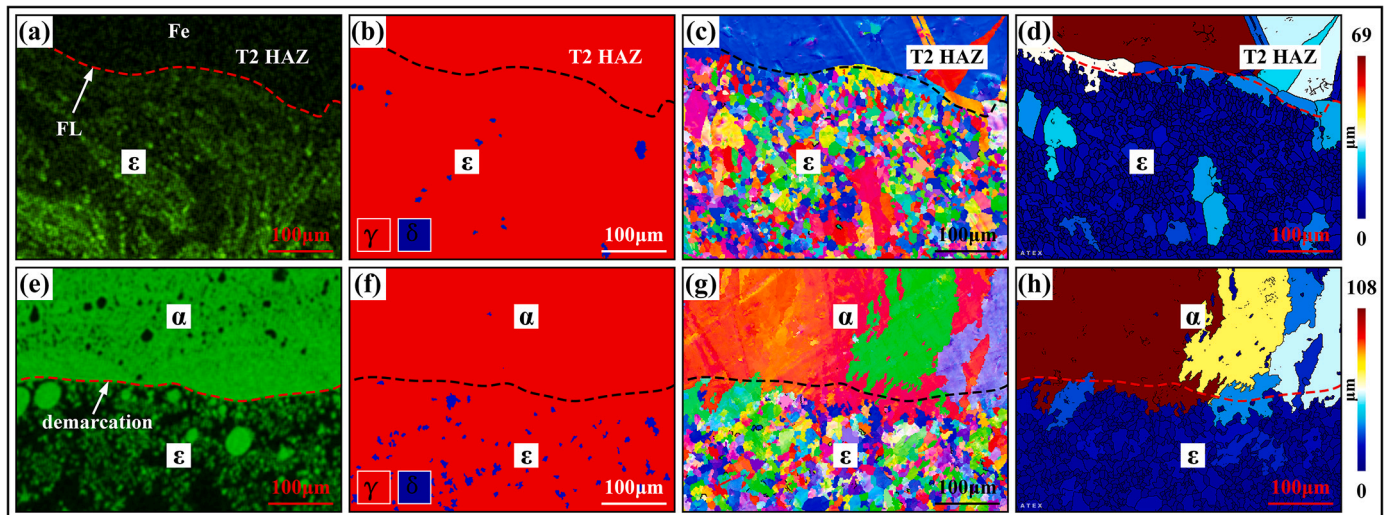


Fig. 9. Fe mapping, phase map, IPF orientation map and grain size map of epitaxial (a-d) and mixed zones (e-h) on T2 side.

Fe, respectively. As can be seen, the former interpreted the formation of Cu rich ϵ and Fe-Cr-Ni rich α structures by first and second liquid phase separations. Moreover, the solidification sequence in the laser welded 304-T2 joint followed $\delta \rightarrow$ Fe-Cr-Ni rich $\delta/\gamma \rightarrow$ Cu rich γ during the

cooling process. According to some previous studies, the previously precipitated phase was supposed to determine or influence the successive grain growth and even the orientation.

Based on the phase transformation and experimental results, the

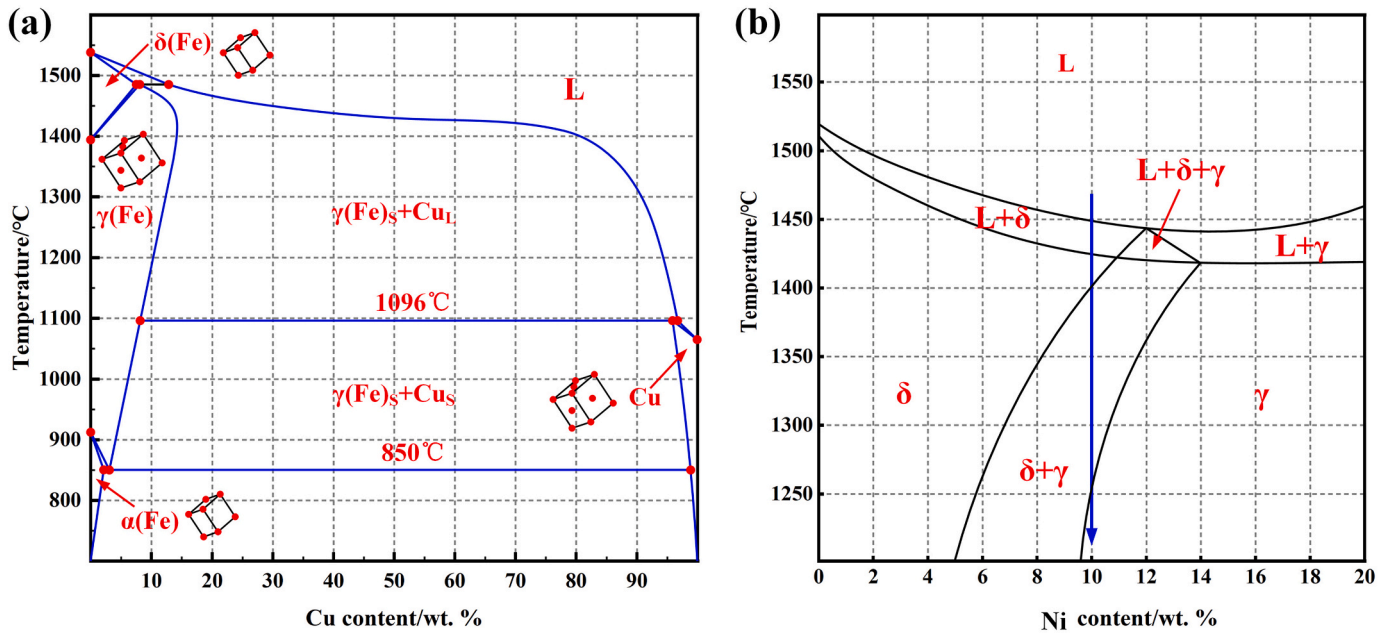


Fig. 10. Phase diagrams of (a) Fe-Cu and (b) Cr-Ni at 70% Fe.

formation process of Cu rich ϵ and Fe-Cr-Ni rich α structures were displayed as Fig. 11 concerning various morphology and grain size profiles. During the formation of 304-T2 welded joint fabricated by deflected laser, resultant macro morphology in the transverse section was composed of three regions based on ϵ arrangement on α matrix. As is divided, region I represented the blocky ϵ without apparent directional attribute. Such region was immediately in the vicinity of T2 FL. Accordingly, the presence of first-to-solidify α spheres in ϵ was doomed to suppress the convection behavior under the high heat diffusion rate. In the region II near the bottom of 304 FL, the relatively low heat diffusion rate was capable of generating enough Marangoni flow and relevantly simultaneous advection of bonded ϵ and inside α spheres. For the last-to-solidify region III, long eduration of liquation situation offered an adequate reconstrastion of ϵ and inside α to present in spherical fashion with low interfacial energy.

On a micro scale, two cases were considered for the localied solidification sequence and final crystal orientation arrangement. During the formation of Cu rich ϵ structure, dispersed Fe-Cr-Ni rich spheres first solidified with random orientations under unstable and fluctuant temperature field. As the residual melt was cooled to Cu solidus, epitaxial

growth happened between solidifying Cu rich γ and solidified Fe-Cr-Ni rich δ/γ . In such course, the predetermined orientation of the latter was succeeded to the former based on the lowered interfacial energy caused by disordered lattice. Resultantly, there exiated a number of high disorientation boundary in Cu rich ϵ . As for Fe-Cr-Ni rich α , Fe-Cr-Ni rich δ/γ also possess a priority by primary solidification. In addition, the minor fraction of liquid Cu melt was unable to disturb the continuous growth and uniform orientation of bulk Fe-Cr-Ni rich δ/γ . In this way, the subsequent solidification of residual Cu rich γ also kept to the bulk Fe-Cr-Ni rich δ/γ , ultimately giving rise to a coincident crystalline texture with few high disorientation. Such difference also interpreted the progressively wider epitaxial zone on 304 side than T2 side.

4. Conclusions

In this paper, the deflected laser welded joint of 304 stainless steel and T2 copper was studied concerning structural arrangement, grain structure and texture evolution on macro and micro scales. Main conclusions were listed as follows:

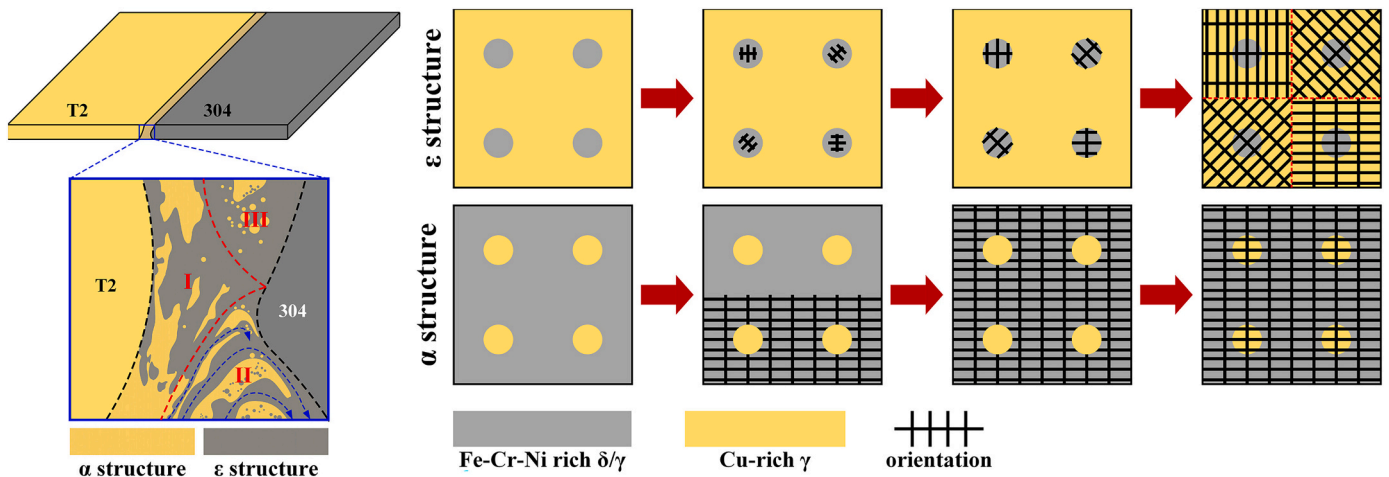


Fig. 11. Formation process and solidification features of Cu-rich ϵ and Fe-Cr-Ni rich α structures on macro and micro scales.

1. The laser welded fusion zone contained two kinds of microstructure including major fraction of Fe-Cr-Ni rich α structure (spherical Cu in Fe matrix) and minor fraction of Cu-rich ε structure (spherical Fe in Cu matrix). Overall, FZ performed as a bridge in 304 BM and T2 BM concerning the microstructure and microhardness. In fusion zone, α possessed a relatively higher hardness than ε .
2. Morphologically, three types of ε were observed including blocky, curved and spherical forms. Texturally, there existed seven regions from 304 side to T2 side containing 304 BM, 304 HAZ, 304 epitaxial zone, mixed zone and T2 epitaxial zone, T2 HAZ and T2 BM. Both HAZs were coarsened while 304 epitaxial zone was progressively wider than the T2. In mixed zone, ε possessed comparably dispersed orientation and fine grains.
3. According to the phase diagram and EBSD analysis, the solidification sequence in the fusion zone was $\delta \rightarrow$ Fe-Cr-Ni rich $\delta/\gamma \rightarrow$ Cu rich γ . In this way, the dispersed distribution of primary solidification of Fe-Cr-Ni rich spheres with random orientation led to the formation of Cu rich fine grains with various orientations in ε . As for α , the continuous growth of Fe-Cr-Ni rich bulks with uniform orientation promoted the subsequent growth of inside Cu rich spheres in an identical orientation.

Data availability statement

The raw/processed data required to reproduce these findings cannot be shared at this time due to legal or ethical reasons.

Declaration of Competing Interest

None.

Acknowledgments

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